
Measurement Systems

An Open-Source Summary

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December 16, 2025

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Note

This summary will only be based on the slides and lectures. No figures or copy from the cursus book is permitted due to copyrights.

Nonetheless, it is recommended to also go through the cursus book as it may go more in depth than this summary

Part I.

Introduction

1. Introduction

1.1. Why is measuring important ?

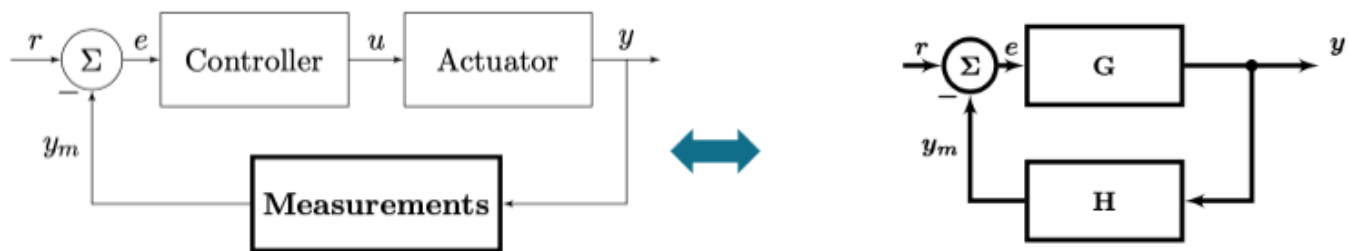


Figure 1: Typical system

In a feedback system, we want some input to transform in a specific output. Later, we want to measure this output to adapt the control and see how effective this is.

Using the standard G and H notation we can find that:

$$y = \frac{G}{1 + GH} r \quad e = \frac{1}{1 + GH} r \quad (1)$$

$$y = \lim_{G \rightarrow \infty} \frac{G}{1 + GH} r \approx \frac{1}{H} r \quad (2)$$

1.2. General principles

Building a measurement system is transforming energy/information from one domain to another.

We first start by building the physical knowledge (build physical quantity model), mathematical reasoning (find how accurate we can make the model to reflect the system) and finally we must take noise and other limits into consideration.

In a system, at a high-level, there is 2 forms of variables:

1. **Across Variables:** describe the *state*, in **parallel** with the terminal. Effort to change the state

2. **Through Variables:** describe the *flow*, in **series** with the terminal.

Then combining those two can give us some fundamental quantities: - $A \cdot T$: **power** generated or dissipated in the element - A/T : **impedance** of the system. Describes how an element transforms an A into a T

1.3. Modelling a system : the physical model

We usually want to map the IO behavior based on physical knowledge. Each model tries to represent reality but will always introduce errors or not take into account higher order phenomena.

For example, we can model a pressure change using a diaphragm to a capacitance change. If we take a simple spring model we don't take second order term into account which introduces errors.

Table 1: Strength and weaknesses of physical approach

Strength	Weakness
Good insight	Error due to approximations
System optimization is possible	Derivation of this formula is not always straightforward/possible
Calibration techniques can be easily derived	Errors due to tolerances in fabrication

1.4. Correlation

A typical operation used in system is **correlation**, it is used to find the “likelihood” between two signals. It helps us extract features:

$$R_{fg}(t) = \int_{-\infty}^{\infty} f(t)g(t - \tau)d\tau$$

Knowing Fourier's theory, we can represent any signal as a combination of *sin* and *cos*. Instead of using $g(t - \tau)$, this gives us:

$$F(f) = F_{\cosine}(f) + F_{sine}(f) \quad (3)$$

$$= \int_{-\infty}^{\infty} (\cos(2\pi ft) + i\sin(2\pi ft))f(t)dt \quad (4)$$

1.4.1. Freshen up of Laplace transform

The Fourier transform can be written as:

$$\mathcal{F}f(t) = \int_{-\infty}^{\infty} e^{i2\pi ft} f(t)dt$$

The Laplace transform goes one step further by using the Euler notation $e^{ix} = \cos(x) + i\sin(x)$ and extend the Fourier transform with complex numbers:

$$\mathcal{L}f(s) = \int_0^{\infty} e^{-st} f(t) dt \quad s = \sigma + i\omega t$$

It can be seen that:

$$\mathcal{F}(f(t)) = \mathcal{L}(f(t)) \iff f(t) = 0 \text{ if } t < 0$$

Laplace transform is an easy tool to solve differential equation and work with derivative:

$$\mathcal{L}\left(\frac{df(t)}{dt}\right) = s\mathcal{L}(f(t)) + f(0)$$

1.5. The Measurement Chain

A measurement system converts a state variable in a measured value

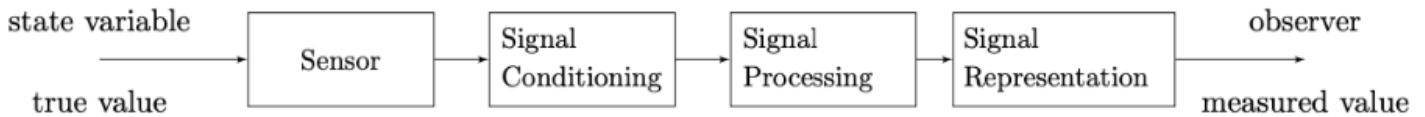


Figure 2: 4 major components

The idea behind a *signal conditioning* is to reduce error and filter out the signal. This should also convert it into a digital signal that can be easily processed by a DSP block. DSP block is used to work with the input and realize complex function while the signal representation can be a screen, plot, ...

1.5.1. 4 major components

1.5.1.1. Sensor Each sensor is made of 2 sensors to be exact.

1. A primary that transform a physical quantity to an “easier to measure” quantity
2. A secondary that converts this quantity into an electrical signal

1.5.1.2. Signal conditioning This is usually an ADC block. Such block are limited by the current technologies and we call that a **technological constant** C_T . There is no free lunch and everything is a tradeoff leading to a basic formula from information theory:

$$\frac{Speed(Accuracy)^2}{Power} = C_T$$

An ADC needs an input **and** a reference voltage. I outputs $\frac{V_{in}}{V_{ref}} = \frac{N}{2^n - 1}$. The flash converter using a resistive ladder is a typical (but bad way) to convert into digital. We also need a clock and comparators to convert.

Accuracy is important and the Effective Number of Bits or ENOB is a crucial metric for any system. Due to quantization error and the SNR required for 1 valid bit, we can find such formula¹ :

¹ Find more information about how we can get such formula in the DAMSIC course, [link to a summary of the course](#). Check the section 4.1.2

$$ENOB = \frac{SNDR - 1.76dB}{6.02dB}$$

So in summary, building a physical models is a good step in the right direction to be able to better filter and analyze measurements.

2. Characterization of measurements

2.1. Measurement Errors

In any system, there will be error which will reduce its accuracy. The error is added after each operation in our model but it is important to understand how it **propagates** and its **origin**.

Table 2: Comparing the two type of errors

Characteristics	Absolute error	Relative error
Math	$e = x_m - x$	$e_r = \frac{e}{x} = \frac{x_m - x}{x}$
Unit	Same unit as variable	Unitless - ppm
Estimation	Over-estimation, $e \propto G$	Under-estimation, $e_r! \propto G$

$$x_m = S(s)G(s)x + G(s)\bar{e}_s + \bar{e}_c \quad \text{Output measurement} \quad \bar{e}_m = G(s)\bar{e}_s + \bar{e}_c \quad \text{Error of this measurement} \quad (5)$$

$$y = \frac{x_m}{S(0)G(0)} \quad \text{Transfer function} \quad \bar{e}_i n = \frac{\bar{e}_s}{S(s)} + \frac{\bar{e}_c}{S(s)G(s)} \quad \text{Input referred noise} \quad (6)$$

Table 3: Type and difference of errors

Type	Effect	Example
Interfering	Additive to the signal	Magnetic interferences
Modifying	Modifies transfer function	Pressure sensor changes over deformation

By understanding the nature of an error, we can reduce it or even make it disappear. Example, EM interference can disrupt ESG signals. Solution, add an extra probe (right leg) to also capture this interference and connect it to the V_{ref} . Now, V_{in} and V_{ref} in the ADC will have the same error which will cancel out.

2.2. Error Propagation

We must understand how an error propagates in a system. We can build a simple model like:

$$x_m = ax + by$$

Where x is the true value that will be measured x_m , but also possible interference such as y . The error is:

$$e_m = a\Delta x + b\Delta y$$

We can also conduct some statistical analysis:

$$\text{Var}(x_m) = \mathbb{E} \left[(ax + by) - (\mathbb{E}(ax + by))^2 \right] \quad (7)$$

$$= \dots \quad (8)$$

$$\sigma_m^2 = a^2\sigma_x^2 + b^2\sigma_y^2 + 2ab\sigma_{xy} \quad (9)$$

The last term is often forgotten because a lot of person assume $\perp$ between x and y which is not always the case. The covariance is null only if uncorrelated. So typically, if the errors are due to the same source, their covariance is most likely correlated.

2.2.1. Non-linear functions

For more advance transfer function that are non-linear, we can rewrite it using the Taylor series:

$$f(x, y) = f(x_1, y_1) + \frac{\partial f}{\partial x}(x - x_1) + \frac{\partial f}{\partial y}(y - y_1) + \dots$$

Assuming a small error we can simplify our function and say that

$$\sigma_m^2 \approx \frac{\partial f^2}{\partial x} \sigma_x^2 + \frac{\partial f^2}{\partial y} \sigma_y^2 + 2 \frac{\partial f}{\partial x} \frac{\partial f}{\partial y} \sigma_{xy}$$

2.2.2. Static Characteristics

- **Range & Span:** range is the min and max while the span is the delta between the two
 - Those values should reflect where the system is still meeting its specifications

2.2.2.1. Linearizing the reading Knowing the range, we could force a simple linearization using:

$$O - O_{min} = \frac{O_{max} - O_{min}}{I_{max} - I_{min}}(I - I_{min})$$

But this is far from ideal, a better solution is the least square solution:

$$K = \frac{Cov(X_1, Y_1)}{Var(X_1)} \quad (10)$$

$$= \frac{X_1 Y_1 - n \bar{X}_1 \bar{Y}_1}{X_1 Y_1 - n \bar{X}_1^2} \quad (11)$$

Table 4: Quick overview of the metrics

Metric	Description	Issue
Max error	Maximum error from ideal and model	Doesn't give a real insight
DNL	Difference between width of the step and ideal LSB step. <1LSB or problematic	Watch out for sign inversion
INL	Maximum deviation (max(DNL))	Less insight
THD	Each non-linearity causes distortion, highlight the distortion nature	Doesn't give information about the non-linearity characteristic

It is important to understand the error of the non-linearity. Because compensating for it is costly as we will use a polynomial to cancel it. The order of the polynomial will require $n + 1$ calibration measurements.

2.2.2.2. Resolution

Definition

Resolution is the smallest discrete step a system can take

$$resolution = \frac{\Delta I_R}{I_{max} - I_{min}} \cdot 100\%$$

The resolution isn't the accuracy as we could add dummy 0 which won't improve the accuracy but the resolution.

2.2.2.3. Hysteresis It is a truly physical phenomena that depicts how when changing the input and reducing it will not lead to the same output path due to various reasons.

2.2.2.4. Dynamical errors When measuring something, we often think we are in the steady state which is not really the case. Most of the time we will wait x amount of time until it is "good enough".

This is all linked with the **memory elements** that is present in mechanical, electrical, ... systems. The more **independent** memory elements we have the higher is the order of the system. We talk here about settling time which models the *exponential* behavior of such event;

$$\varepsilon = \Delta I_{expt} \left(-\frac{t}{\tau} \right)$$

Of course, Taylor series can be used for small amplitude: $TF(s) = \frac{\partial O}{\partial I}|_{I=I^*} 1/(1 + s\tau)$

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2.3. Measurement Characteristics

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2.3.1. Quantization noise

When measuring through an ADC, we can see the loss of precision due to limited bitwidth as noise. It is not physical noise but more like a mathematical tricks to model the impact of quantization on signal. For this to work we must use the postula that the input is a varying signal. If it is not a varying signal, the error will be constant and not looking like white noise aka spread equally. The mean value of this noise will be 0:

$$\mathbb{E}(e_q) = \frac{1}{\Delta I_R} \int_{-\frac{\Delta I_R}{2}}^{\frac{\Delta I_R}{2}} e \, de = 0$$

The power can be seen as:

$$\sigma_q^2 = \frac{1}{\Delta I_R} \mathbb{E} \left((e_q - \mathbb{E}(e_q))^2 \right) \quad (12)$$

$$= \frac{1}{\Delta I_R} \int_{-\frac{\Delta I_R}{2}}^{\frac{\Delta I_R}{2}} e^2 \, de \quad (13)$$

$$= \frac{\Delta^2 I_R}{12} \quad (14)$$

As used and explained earlier, the σ^2 is the power of the signal (total integrated noise). The spectrum is limited to $f_s/2$. So we can model an ADC with an extra white noise source with a noise power of $\frac{\Delta^2 I_R}{12}$ with a bw of $f_s/2$.

3. Building a chain : reducing errors

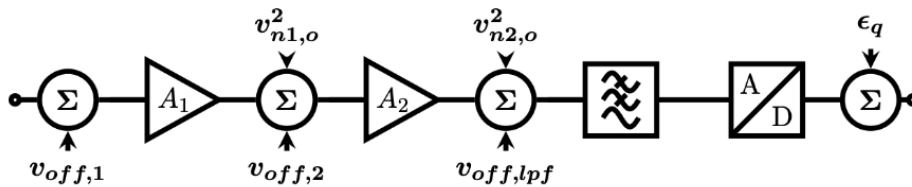


Figure 3: Measurement chain

Each blocks add offset, gain error and noise. This is commonly represented as the chain above.

3.1. Gain distribution

A common way to summarize a chain is by using *input referred* value. This also showcases some basic idea in measurement chains.

$$v_{in}^2 = \frac{v_{n1,o}^2}{A_1^2} + \frac{v_{n2,o}^2}{A_1^2 A_2^2} \quad I_{ADC} = (V_{in} + v_{off,1})A_1 A_2 + v_{off,2}A_2$$

We have to maximize the first stage gain as it will reduce the impact of noise for later stage components. But we are limited by the offset in the ADC.

3.2. Noise filtering

Another way to reduce the noise by averaging it:

$$\sigma_n^2 = \sum_i^N \frac{x_i - \bar{x}}{N - 1}$$

This corresponds to filtering in the frequency domain. We have to be careful with aliasing and this technique cannot refer the $1/f$ noise.

3.3. Chopping and modulation

Offset and $1/f$ cannot be filtered sadly. We would need 0 bandwidth filter for this. Another solution would be to **shift** the noise to higher frequencies:

$$\sin(\omega t) \sin(\omega_c t) = \frac{\cos((\omega_c - \omega)t) + \cos((\omega_c + \omega)t)}{2}$$

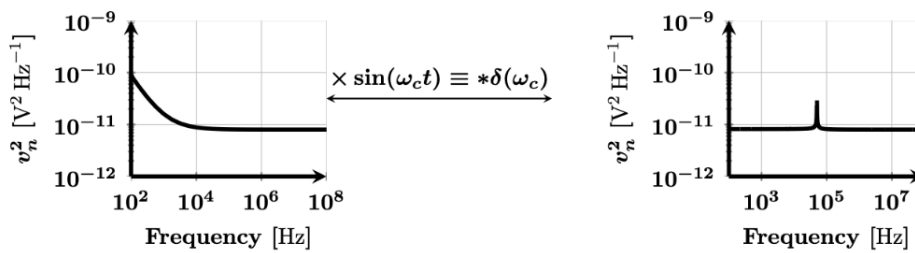


Figure 4: Chopping

3.3.1. Demodulation

Using a sine, half of the energy is lost sadly. This is why actual chopping with a square wave is better.

3.3.2. Chopping

$$S(t)S(t)|_{\omega=0} = \left(\frac{4}{\pi}\right)^2 \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} \sin\left(2\pi(2n+1)\frac{t}{T}\right) \sin\left(2\pi(2n+1)\frac{t}{T}\right) = \frac{8}{\pi^2} \frac{\pi^2}{8} = 1$$

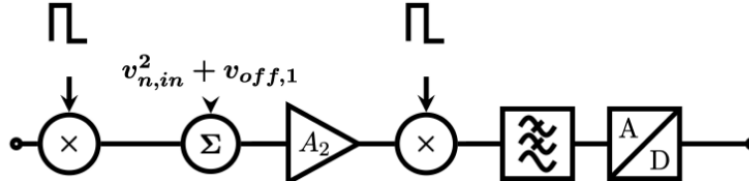


Figure 5: Chopping chain - with representation of the tones (no need to remember the math)

This will retain the signal energy. The input signal is modulated with pulse train, offset + $1/f$ added by A_2 . Then demodulated at higher frequencies. Thus, a low-pass filter will not take the high frequency $1/f$ and offset. We are using a 1 -1 square wave.

We must have perfect reconstruction but amplifier has a finite bw. This cannot be easily realized as real components are not perfect. Chopping error.

The square wave signal contains all of the frequency and some beyond the amplifier A_2 . This result also in chopping spikes. The chopping error at $2\omega_c$ will be gone but still problems. Energy put at $2\omega_c$ is also lost at the 0 frequency. It is a gain error, if 0 at input, 0 at output no error. The stronger is the signal the larger is the error thus gain error. The error becomes:

$$\frac{4\pi}{\sqrt{2}} \sqrt{\left(\frac{\tau}{T}\right)^2 \frac{1}{1 + \frac{\tau}{T}}}$$

3.4. Feedback

A good solution is feedback as it will remove distortion and improve settling.

$$y = \frac{MG}{1 + MGH} V_{in} + \frac{G}{1 + MGH} \varepsilon \approx \frac{V_{in}}{H} + \frac{\varepsilon}{MH} = \frac{V_{in}}{H} \quad G \rightarrow \infty$$

3.4.0.1. Example Measurement of current in electric car. Use a TMR system but has 2 issues:

1. hysteresis
2. temperature issue: modifying environment error

We cannot directly measure on the path, add coils to counter-act the magnetic field which will linearize the reading by fighting an EM field effect with another EM field.

3.5. Calibration

Definition

Calibration is a method to remove an error by measuring the signal with a better sensor and using this measurement to compensate errors (typically drift)

Typically realize offset and gain calibration. This can be realized in the factory, online (using a switch that measure known value then compare. Can also be a random signal in more advanced system) or timed calibration.

If we are measuring and calibrating with a worse system, we are increasing our error.

3.5.0.1. Gain error example

$$V_{out,cal} = (BS + V_{offset} - a)b \quad \hat{a} = b(a - V_{offset}) \quad (15)$$

$$V_{out,cal} = B\hat{b} + \hat{a} \quad \hat{b} = Sb \quad (16)$$

$$\sigma_{\hat{b}}^2 = \left(\frac{\partial \hat{b}}{\partial B_1} \right)^2 \sigma_{B_1}^2 + \left(\frac{\partial \hat{b}}{\partial B_2} \right)^2 \sigma_{B_2}^2 + 2 \left(\frac{\partial \hat{b}}{\partial B_1} \right) \left(\frac{\partial \hat{b}}{\partial B_2} \right) \sigma_{B_1 B_2} + \sigma_{V_{out}}^2$$

Error on the applied field

Gain error of the calibration equipment

Noise of the calibration equipment

Figure 6: The calibration is as good as the calibration equipment

We need two measurements to be able to model the error. We can see in the math that our calibration will be as good as the accuracy and noise we have.

Need long time to measure to average out the noise. Know the curve and what we should be expecting. Spread the points as wide as possible. A small error between two close points will result in large issues. Calibration is thus a mean to remove *stochastic error*.

3.6. Compensation**Definition**

Compensation is a technique where an environmental error is removed by measuring this variable using another sensor

3.6.1. Hall sensor compensation

Typically, if we know that a sensor may be subjected to variation due to temperature, we will try to measure temperature too to counteract this effect.

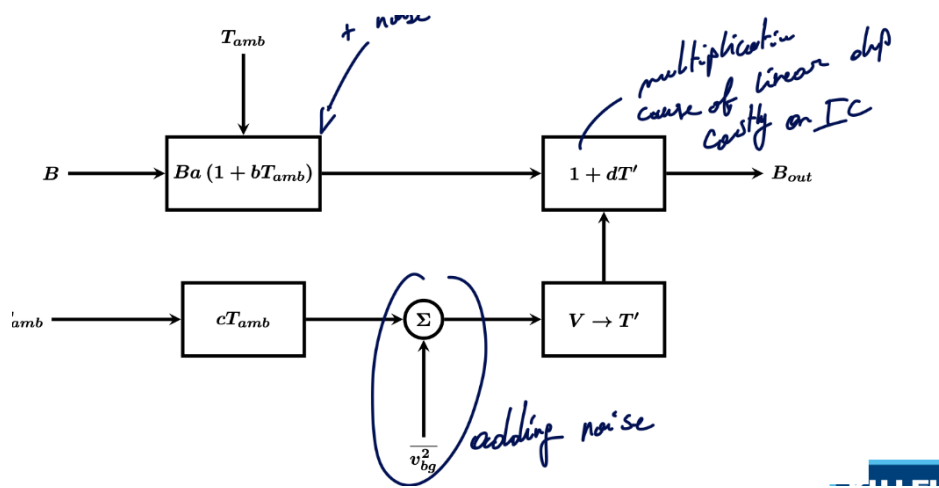


Figure 7: Compensation of a Hall sensor

Note

Here, we multiply T' by d instead of dividing it as division is an expensive operation in digital.

The output of this chain is:

$$B_{out} = Ba(1 + bT_{amb} + cdT_{amb} + bcdT_{amb}^2)$$

We want the B_{out}/B ratio to be constant, in other words not dependent on the temperature. One solution is to make the slope 0 at some temperature. This will require **2** calibration points.

We could also use a quadratic measurement chain. This can make the slope equal 0 on a larger interval of temperature.

3.6.1.1. noise of the temperature sensor But, we will also have noise coming from this compensation chain. Namely the added noise is modeled as:

$$\sigma_{B_{out}}^2 = \left(\frac{dB_{out}}{dT'} \right)^2 \bigg|_{T'=T_{amb}} \sigma_{T'}^2$$

Luckily, the temperature is a slow varying process meaning that the bandwidth is in the Hz range. This can be thus filtered out as magnetic field has a bandwidth in the kHz range.

3.7. Conclusion

- White noise can be filtered
- Flicker noise limits the applicability of filtering
 - Move $1/f$ to higher frequencies : chopping
- Static errors can be removed using calibration
- Environmental errors can be removed using compensation
- BUT : every correction technique adds errors

- Filtering → settling error
- Chopping → gain error
- Calibration → as good as calibration equipment
- Compensation → add noise of compensating sensor

4. Measuring electrical quantities

4.1. OpAmp

Basic facts about OpAmp:

- High input impedance: so almost no current flows in
- Low output impedance: delivers lots
- Very high gain
- Has a positive and negative terminal

- Non-inverting amplifier

$$v_{fb} = \frac{R_1}{R_1 + R_2} v_{out}$$

$$A(v_{in} - v_{fb}) = v_{out}$$

$$\frac{v_{out}}{v_{in}} = \frac{A(R_1 + R_2)}{R_1 + R_2 + AR_1}$$

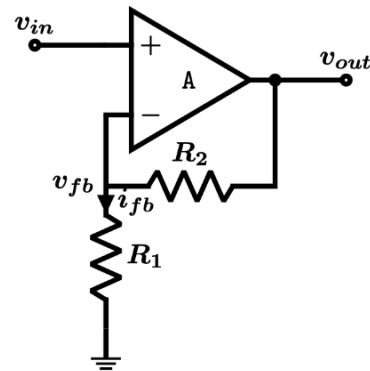


Figure 8: Non-inverting amplifier

With $\lim_{A \rightarrow \infty}$ we get $1 + \frac{R_2}{R_1}$. This means we can control the exact gain by controlling the resistance ratio. Then, by combining two together, we can make an *instrumentation amplifier*:

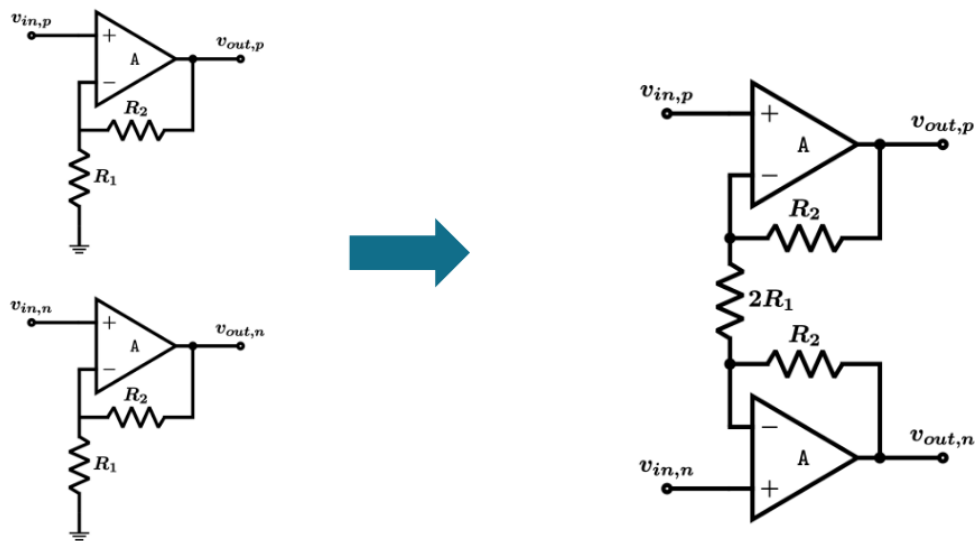


Figure 9: Instrumentation amplifier

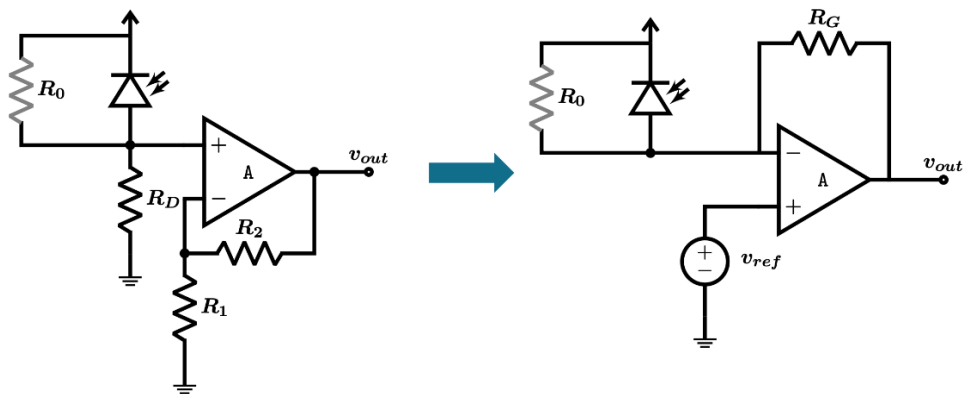


Figure 10: Measuring the current

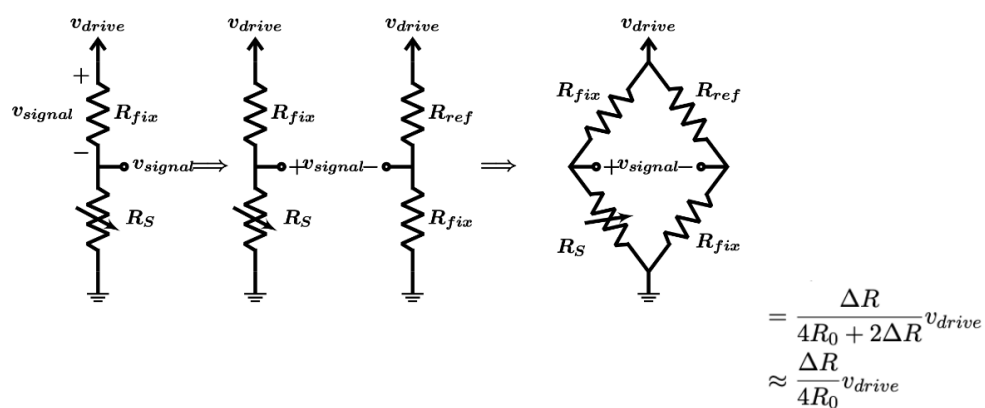


Figure 11: Wheatstone bridge

In the Wheatstone bridge, the $2\Delta R$ factor at the denominator will create non-linearity.

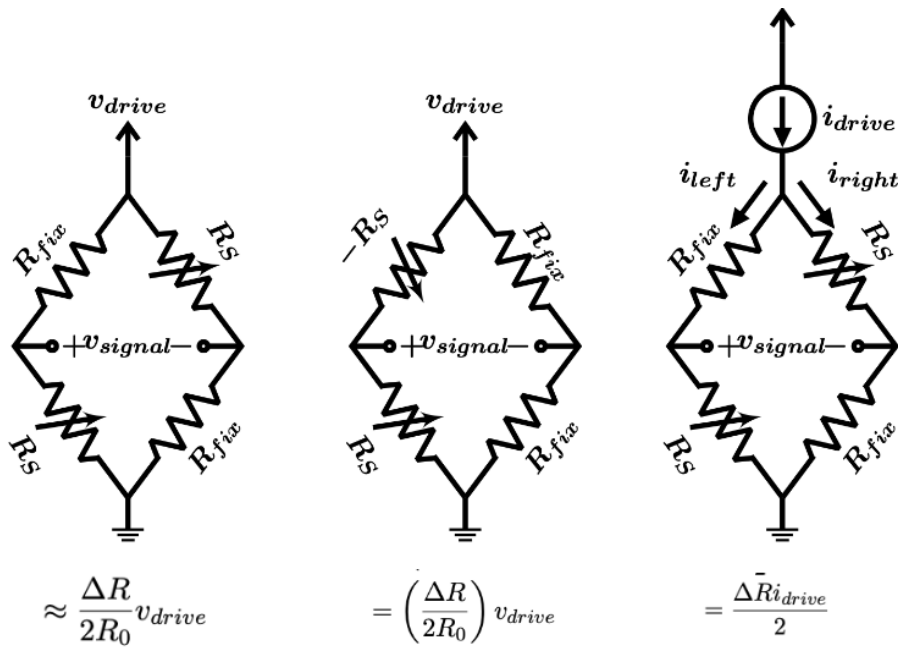


Figure 12: Improvements

4.2. Measuring reactive elements

As seen previously, reactive elements cannot be measured with DC currents as they will either short or be open. A good way is to apply a chopping/square signal instead of a $\sin$ to avoid the reduction of energy by a factor 2.

4.2.1. Using oscillators to measure reactive elements

An oscillator is a feedback device that becomes a positive feedback. This must respect the **Barkhausen criterium**:

$$|G(j\omega_n)H(j\omega_n)| = 1 \quad (17)$$

$$\angle G(j\omega_n) + \angle H(j\omega_n) = \pi \quad (18)$$

The π indicates an inverse of the sign. Instead of a negative feedback, it becomes positive.

- $$H(s) = \frac{1}{s^2 LC/R + sL/R + 1}$$

- This gives as criterion for the opamp

$$|G(j\omega_n)| = 2\zeta$$

$$\angle G(j\omega_n) = -\pi - \angle H(j\omega_n) = -\frac{\pi}{2}$$

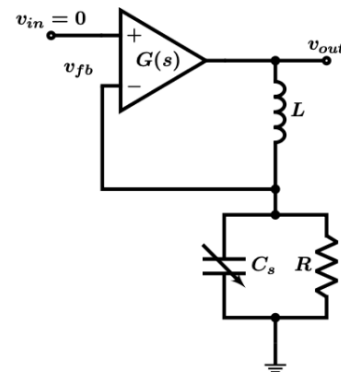


Figure 13: Measuring reactive elements

The resistance must be compensated.

$$\omega_n = \frac{1}{\sqrt{LC}} \quad \zeta = \frac{1}{2R} \sqrt{\frac{L}{C}} \quad Q = R \sqrt{\frac{C}{L}}$$

The Q factor is a metric that depicts the quality of a system. It is the ratio of the total energy of the system over the power dissipated over a cycle ω :

$$Q = \frac{1}{2\zeta} = \omega \frac{\text{Max energy stored}}{\text{Power loss}}$$

The higher is the Q factor, the narrower is the oscillation band. This makes it tough to oscillate but once in oscillation, it is hard to stop it.

4.2.1.1. Measuring the frequency We can use a counter based on a system clock that will monitor the oscillation. The precision of this measurements is as good as the digital clock (jitter, noise, ...). Of course, the clock should have a higher frequency than the frequency we are trying to monitor (Shanon sampling frequency, ...).

$$\Delta\omega = \left(\frac{\partial\omega_n}{\partial C} \right) \Delta C \quad (19)$$

$$= \frac{1}{2} \frac{1}{\sqrt{LC^3}} \Delta C \quad (20)$$

$$= \frac{\omega_n}{2C_0} \Delta C \quad (21)$$

4.3. Conclusion

- The design of a measurement chain is aimed to reduce the errors :
 - Meet the input range of the ADC to reduce quantization noise
 - * to avoid clipping or simply not using the full dynamic range
 - Filter of noise
 - Use chopping / modulation to be able to split frequencies
 - Calibration can be used to reduce production errors
 - * For stochastic
 - Compensation to reduce environmental errors
 - Use feedback to reduce errors and improve dynamics

Part II.

sensors and specific measurement equipment

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